Zhaoxuan Tan

Ph.D. Candidate, Computer Science and Engineering, University of Notre Dame

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Research Interests

My research focuses on building personalized foundation models and intelligent systems that understand and adapt to each individual. I study how models infer people's preferences from implicit signals such as user-generated content and interaction histories, and how to adapt models to each user in modular, efficient, and scalable ways. I am also interested in personalized agents with long-term memory that proactively assist individuals and groups.

Education

University of Notre Dame, Notre Dame, IN Aug 2023 – May 2027 (expected)
Ph.D. in Computer Science and Engineering; advisor: Prof. Meng Jiang
Thesis: *Building Personalized Foundation Models and Intelligent Systems*

Xi'an Jiaotong University, Xi'an, China Aug 2019 – Jul 2023
B.E. in Computer Science and Technology; advisor: Prof. Minnan Luo

Industry Research Experience

Microsoft, Office of Applied Research, Redmond, WA May 2026 – Present
Research Intern; hosts: Dr. Pei Zhou, Dr. Mengting Wan

- Developing shared-context briefs for proactive agents in long-term, multi-person Teams collaboration, distinguishing decisions from proposals while preserving attribution, unresolved disagreements, and evolving project state. A version of the briefer is already in use internally.
- Evaluating claim grounding, attribution, and freshness using GroupMemBench-based simulations with oracle references; early results show stronger recovery of past state than prediction of future opportunities and risks.

Google, CoreML, Remote Oct 2025 – Apr 2026
Student Researcher; hosts: Dr. Yan Zhu, Dr. Xiang Zhai

- Developed BLUE, a reinforcement learning framework that trains an LLM to generate interpretable user profiles using recommendation rewards and a next-item prediction objective.
- Improved recommendation across five domains with a frozen embedding model, including zero-shot transfer from Amazon Clothing to three Amazon categories and Google Local Review. First-author paper: NeurIPS 2026.

Amazon, Rufus, Palo Alto, CA Jun 2025 – Oct 2025
Applied Scientist Intern; hosts: Dr. Zixuan Zhang, Dr. Haoyang Wen

- Developed Profile-to-PEFT, an end-to-end hypernetwork that maps user profiles directly to LoRA adapters, eliminating per-user training at deployment and generalizing to unseen users.
- Outperformed prompt-based personalization and per-user fine-tuning (OPPU); generated personalized adapters in 0.57 s per user, achieving 33× faster adaptation than the fastest OPPU variant. First-author paper: ACL 2026.

Microsoft, Office of Applied Research, Remote Mar 2025 – Jun 2025
Research Intern (part-time); hosts: Dr. Pei Zhou, Dr. Mengting Wan

- Built the Meeting Dynamic Forecasting Benchmark (MDFB) from 2,207 real-world meetings and 24,794 forward-looking queries to evaluate LLMs' ability to forecast multi-party discussions, tracking agendas, roles, and disagreement.
- Contributed to GLARE, an adversarial-reward post-training method achieving average human-evaluated win rates of 66% on utility and 70% on human-likeness across model and human baselines.

Amazon, Rufus, Palo Alto, CA May 2024 – Oct 2024
Applied Scientist Intern; hosts: Dr. Zheng Li, Dr. Tianyi Liu

- Developed PUGC, converting unlabeled user-generated content into scalable preference data for DPO.
- Achieved a 35.93% length-controlled win rate on AlpacaEval 2.0 with Mistral-7B-Instruct-v0.2, 9.37 percentage points above DPO with UltraFeedback preference data. First-author paper: ACL 2025.

Publications

* denotes equal contribution. Citation metrics and updates: Google Scholar.

First-Author Publications

[1] **Bridging Textual Profiles and Latent User Embeddings for Personalization.**
Zhaoxuan Tan, Xiang Zhai, Yan Zhu, Meng Jiang, Mohamed Hammad. *NeurIPS 2026.*

[2] **Instant Personalized Large Language Model Adaptation via Hypernetwork.**
Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, Rongzhi Zhang, Pei Chen, Fengran Mo, Zheyuan Liu, Qingkai Zeng, Qingyu Yin, Meng Jiang. *ACL 2026.*

[3] **Aligning Large Language Models with Implicit Preferences from User-Generated Content.**
Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, Hyokun Yun, Ming Zeng, Pei Chen, Zhihan Zhang, Yifan Gao, Ruijie Wang, Priyanka Nigam, Bing Yin, Meng Jiang. *ACL 2025.*

[4] **Can Large Language Models Understand Preferences in Personalized Recommendation?**
Zhaoxuan Tan, Zinan Zeng, Qingkai Zeng, Zhenyu Wu, Zheyuan Liu, Fengran Mo, Meng Jiang. *arXiv preprint 2025.*

[5] **Personalized Pieces: Efficient Personalized Large Language Models through Collaborative Efforts.**
Zhaoxuan Tan, Zheyuan Liu, Meng Jiang. *EMNLP 2024.*

[6] **Democratizing Large Language Models via Personalized Parameter-Efficient Fine-tuning.**
Zhaoxuan Tan, Qingkai Zeng, Yijun Tian, Zheyuan Liu, Bing Yin, Meng Jiang. *EMNLP 2024.*

[7] **User Modeling in the Era of Large Language Models: Current Research and Future Directions.**
Zhaoxuan Tan, Meng Jiang. *IEEE Data Engineering Bulletin 2023.*

[8] **BotPercent: Estimating Bot Populations in Twitter Communities.**
Zhaoxuan Tan*, Shangbin Feng*, Melanie Sclar, Herun Wan, Minnan Luo, Yejin Choi, Yulia Tsvetkov. *Findings of EMNLP 2023.*

[9] **KRACL: Contrastive Learning with Graph Context Modeling for Sparse Knowledge Graph Completion.**
Zhaoxuan Tan, Zilong Chen, Shangbin Feng, Qingyue Zhang, Qinghua Zheng, Jundong Li, Minnan Luo. *The Web Conference (WWW) 2023.*

[10] **TwiBot-22: Towards Graph-Based Twitter Bot Detection.**
Shangbin Feng*, Zhaoxuan Tan*, Herun Wan*, Ningnan Wang*, Zilong Chen*, Binchi Zhang*, ..., Jundong Li, Minnan Luo. *NeurIPS 2022, Datasets and Benchmarks Track.*

Co-Authored Publications

[11] **Prompt-Activation Duality: Improving Activation Steering via Attention-Level Interventions.**
Diancheng Kang, Zheyuan Liu, Ningshan Ma, Yue Huang, Zhaoxuan Tan, Meng Jiang. *NeurIPS 2026.*

[12] **Steering Multimodal Large Language Models Decoding for Context-Aware Safety.**
Zheyuan Liu, Zhangchen Xu, Guangyao Dou, Xiangchi Yuan, Zhaoxuan Tan, Radha Poovendran, Meng Jiang. *EMNLP 2026.*

[13] **DRIFT: Learning from Abundant User Dissatisfaction in Real-World Preference Learning.**
Yifan Wang, Bolian Li, Junlin Wu, Zhaoxuan Tan, Zheli Liu, Ruqi Zhang, Ananth Grama, Qingkai Zeng. *ICLR 2026.*

[14] **From Personal to Collective: On the Role of Local and Global Memory in LLM Personalization.**
Zehong Wang, Junlin Wu, Zhaoxuan Tan, Bolian Li, Xianrui Zhong, Zheli Liu, Qingkai Zeng. *Findings of ACL 2026.*

[15] **Agentic Conversational Search with Contextualized Reasoning via Reinforcement Learning.**
Fengran Mo, Yifan Gao, Sha Li, Hansi Zeng, Xin Liu, Zhaoxuan Tan, Xian Li, Jianshu Chen, Dakuo Wang, Meng Jiang. *Findings of ACL 2026.*

[16] **MTMCS-Bench: Evaluating Contextual Safety of Multimodal Large Language Models in Multi-Turn Dialogues.**
Zheyuan Liu, Dongwhi Kim, Yixin Wan, Xiangchi Yuan, Zhaoxuan Tan, Fengran Mo, Meng Jiang. *Findings of ACL 2026.*

[17] **Generalizable LLM Learning of Graph Synthetic Data with Post-training Alignment.**
Yizhuo Zhang*, Heng Wang*, Shangbin Feng*, Zhaoxuan Tan, Xinyun Liu, Yulia Tsvetkov. *Findings of ACL 2026.*

[18] **GLARE: Generative Learning via Adversarial Reward Estimation for Social Dynamics Forecasting.**
Tenghao Huang, Zhaoxuan Tan, Muhao Chen, Jonathan May, Mengting Wan, Longqi Yang, Pei Zhou, Sihao Chen. *arXiv preprint 2026.*

[19] **MoCo: A One-Stop Shop for Model Collaboration Research.**
Shangbin Feng*, Yuyang Bai*, Ziyuan Yang*, Yike Wang, Zhaoxuan Tan, Jiajie Yan, Zhenyu Lei, Wenxuan Ding, Weijia Shi, ..., Luke Zettlemoyer, Yejin Choi, Yulia Tsvetkov. *arXiv preprint 2026.*

[20] **Adaptive Personalized Conversational Information Retrieval.**
Fengran Mo, Yuchen Hui, Yuxing Tian, Zhaoxuan Tan, Chuan Meng, Zhan Su, Kaiyu Huang, Jian-Yun Nie. *CIKM 2025.*

[21] **Enhancing Mathematical Reasoning in LLMs by Stepwise Correction.**
Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, Zhaoxuan Tan, Chao Shen, Meng Jiang. *ACL 2025.*

[22] **Modality-Aware Neuron Pruning for Unlearning in Multimodal Large Language Models.**
Zheyuan Liu, Guangyao Dou, Xiangchi Yuan, Chunhui Zhang, Zhaoxuan Tan, Meng Jiang. *ACL 2025.*

[23] **CodeTaxo: Enhancing Taxonomy Expansion with Limited Examples via Code Language Prompts.**
Qingkai Zeng, Yuyang Bai, Zhaoxuan Tan, Zhenyu Wu, Shangbin Feng, Meng Jiang. *Findings of ACL 2025.*

[24] **IHEval: Evaluating Language Models on Following the Instruction Hierarchy.**
Zhihan Zhang, Shiyang Li, Zixuan Zhang, Xin Liu, Haoming Jiang, Xianfeng Tang, Yifan Gao, Zheng Li, Haodong Wang, Zhaoxuan Tan, Yichuan Li, Qingyu Yin, Bing Yin, Meng Jiang. *NAACL 2025.*

[25] **Protecting Privacy in Multimodal Large Language Models with MLLMU-Bench.**
Zheyuan Liu, Guangyao Dou, Mengzhao Jia, Zhaoxuan Tan, Qingkai Zeng, Yongle Yuan, Meng Jiang. *NAACL 2025.*

[26] **Empirical Guidelines for Deploying LLMs onto Resource-constrained Edge Devices.**
Ruiyang Qin, Dancheng Liu, Zheyu Yan, Zhaoxuan Tan, Zixuan Pan, Zhenge Jia, Meng Jiang, Ahmed Abbasi, Jinjun Xiong, Yiyu Shi. *ACM Transactions on Design Automation of Electronic Systems 2025.*

[27] **KGQuiz: Evaluating the Generalization of Encoded Knowledge in Large Language Models.**
Yuyang Bai*, Shangbin Feng*, Vidhisha Balachandran, Zhaoxuan Tan, Shiqi Lou, Tianxing He, Yulia Tsvetkov. *The Web Conference (WWW) 2024, Oral.*

[28] **Large Language Models Can Self-Correct with Key Condition Verification.**
Zhenyu Wu, Qingkai Zeng, Zhihan Zhang, Zhaoxuan Tan, Chao Shen, Meng Jiang. *EMNLP 2024.*

[29] **Can LLM Graph Reasoning Generalize beyond Pattern Memorization?**
Yizhuo Zhang*, Heng Wang*, Shangbin Feng*, Zhaoxuan Tan, Xiaochuang Han, Tianxing He, Yulia Tsvetkov. *Findings of EMNLP 2024.*

[30] **Chain-of-Layer: Iteratively Prompting Large Language Models for Taxonomy Induction from Limited Examples.**
Qingkai Zeng*, Yuyang Bai*, Zhaoxuan Tan, Shangbin Feng, Zhenwen Liang, Zhihan Zhang, Meng Jiang. *CIKM 2024.*

[31] **What Does the Bot Say? Opportunities and Risks of Large Language Models in Social Media Bot Detection.**
Shangbin Feng, Herun Wan, Ningnan Wang, Zhaoxuan Tan, Minnan Luo, Yulia Tsvetkov. *ACL 2024.*

[32] **DELL: Generating Reactions and Explanations for LLM-Based Misinformation Detection.**
Herun Wan*, Shangbin Feng*, Zhaoxuan Tan, Heng Wang, Yulia Tsvetkov, Minnan Luo. *Findings of ACL 2024.*

[33] **Towards Safer Large Language Models through Machine Unlearning.**
Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, Meng Jiang. *Findings of ACL 2024.*

[34] **Knowledge Crosswords: Geometric Knowledge Reasoning with Large Language Models.**
Wenxuan Ding*, Shangbin Feng*, Yuhan Liu, Zhaoxuan Tan, Vidhisha Balachandran, Tianxing He, Yulia Tsvetkov. *Findings of ACL 2024.*

[35] **LMBot: Distilling Graph Knowledge into Language Model for Graph-less Deployment in Twitter Bot Detection.**
Zijian Cai, Zhaoxuan Tan, Zhenyu Lei, Zifeng Zhu, Hongrui Wang, Qinghua Zheng, Minnan Luo. *WSDM 2024.*

[36] **Machine Unlearning in Generative AI: A Survey.**
Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, Meng Jiang. *arXiv preprint 2024.*

[37] **Can Language Models Solve Graph Problems in Natural Language?**
Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, Yulia Tsvetkov. *NeurIPS 2023, Spotlight.*

[38] **Detecting Spoilers in Movie Reviews with External Movie Knowledge and User Networks.**
Heng Wang, Wenqian Zhang, Yuyang Bai, Zhaoxuan Tan, Shangbin Feng, Qinghua Zheng, Minnan Luo. *EMNLP 2023.*

[39] **KALM: Knowledge-Aware Integration of Local, Document, and Global Contexts for Long Document Understanding.**
Shangbin Feng, Zhaoxuan Tan, Wenqian Zhang, Zhenyu Lei, Yulia Tsvetkov. *ACL 2023.*

[40] **BotMoE: Twitter Bot Detection with Community-Aware Mixtures of Modal-Specific Experts.**
Yuhan Liu, Zhaoxuan Tan, Heng Wang, Shangbin Feng, Qinghua Zheng, Minnan Luo. *SIGIR 2023.*

[41] **HOFA: Twitter Bot Detection with Homophily-Oriented Augmentation and Frequency Adaptive Attention.**
Sen Ye, Zhaoxuan Tan, Zhenyu Lei, Ruijie He, Hongrui Wang, Qinghua Zheng, Minnan Luo. *arXiv preprint 2023.*

[42] **PAR: Political Actor Representation Learning with Social Context and Expert Knowledge.**
Shangbin Feng, Zhaoxuan Tan, Zilong Chen, Peisheng Yu, Qinghua Zheng,
Xiaojun Chang, Minnan Luo. *EMNLP 2022.*

[43] **Heterogeneity-Aware Twitter Bot Detection with Relational Graph Transformers.**
Shangbin Feng, Zhaoxuan Tan, Rui Li, Minnan Luo. *AAAI 2022.*

Honors, Awards, and Research Grants

Tinker Research Grant, Thinking Machines Lab ($5,000 in Tinker API credits)	2026
Lambda Research Grant, Lambda ($5,000 in compute credits)	2026
OpenAI Researcher Access Program, OpenAI ($10,000 in API credits)	2025
Top Reviewer, NeurIPS	2024
Excellent Bachelor's Thesis (ranked 1/172), Xi'an Jiaotong University	2023
AAAI Student Scholarship	2022
National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM)	2021
Dean's List, Xi'an Jiaotong University	2020 – 2022

Professional Service

- Senior Program Committee Member: AAAI 2027.
- Area Chair: ACL Rolling Review (2025–present).
- Conference Reviewer: NeurIPS (2023–; Datasets & Benchmarks 2022), ICLR (2024–), ICML (2025–), ACL Rolling Review (2023–), COLM (2024–), KDD (2024–), AISTATS (2025–), The Web Conference (2024–2025), ICWSM (2023–), Learning on Graphs (2022–2024).
- Journal Reviewer: TMLR, IEEE TKDE, IEEE TNNLS, IEEE/ACM Transactions on Networking.
- Workshop Reviewer: ACL SRW 2025; ICLR AGI 2024; EMNLP WiNLP 2024; ACL KnowledgeNLP 2024; AAAI GCLR 2024; NeurIPS Temporal Graph Learning 2023.
- Volunteer: EMNLP 2022, 2023.

Mentorship and Leadership

Director, LUD Lab, Xi'an Jiaotong University 2022 – 2023
Led a 17-person undergraduate research group.

Selected Mentees (research contributions and subsequent positions, where applicable):

- Heng Wang: NLGraph (NeurIPS 2023 Spotlight), MVSD (EMNLP 2023) → Ph.D., UIUC
- Yuhan Liu: BotMoE (SIGIR 2023) → Ph.D., NYU
- Yuyang Bai: KGQuiz (WWW 2024 Oral) → Ph.D., Texas A&M
- Zijian Cai: LMBot (WSDM 2024) → M.S., Institute of Automation, CAS
- Sen Ye: HOFA → Ph.D., Peking University
- Thomas Mitchell: undergraduate researcher, University of Notre Dame

Teaching

University of Notre Dame

Teaching Assistant, CSE 40113: Design and Analysis of Algorithms	Spring 2024
Teaching Assistant, CSE 30124: Introduction to Artificial Intelligence	Fall 2023